MILITARY SPECIFICATION SHEET

RECEIVERS, RANGE SAFETY, COMMAND
TYPE I, CLASS 1, STYLE A1

This specification sheet forms a part of Military Specification MIL-R-87954A(TE), dated 3 November 1978.

The complete requirements for procuring the receiver described herein shall consist of this document and the latest issue of Specification MIL-R-87954(TE).

REQUIREMENTS:

Electrical:
- Operating frequency (3.5.1.2) and (6.2)
- RF input level monitor (3.5.1.3):
  - DC voltage range: 0 to 4.0 volts or greater, 5.0 volts maximum
  - Output impedance: Less than 5K ohms
  - Rate of change: 0.02 volt/db minimum
  - RF level range: -107 to -80 dbm (min)
  - Repeatability: ±2 db when calibrated at a given temperature from -40°C to 71°C
  - Output connection: Pin A connector J-4, figure 2
  - Load impedance: 47K ohm ± 10%
- Audio output (3.5.1.4): RF input level: > -70 dbm, modulation: ± 30 kHz (± 10%) peak deviation
  - Level: 0.5 volt RMS ± 3 db
  - Frequency range: 7.5 kHz to 32 kHz
  - Load impedance: 47 kilohms
  - Output connection: Pin H J-4, figure 2
- Predetection bandwidth (3.8.1):
  - Reference level: -110 dbm
  - 3 db bandwidth: 180 kHz minimum
  - 60 db bandwidth: 600 kHz maximum
- Decoder command channels (3.5.2):
  - Total number: 4 channels
  - Frequency: (6.2)
- Output section (3.5.3):
  - Output relays (3.5.2.2): Hermetically sealed (double pole/double throw) relay for each command channel to function in accordance with Table 1 Function Table, with RF input level < -110 dbm.
  - Channel status monitor (3.5.2.3): Normally closed contact on 2nd channel, see figure 2.
  - Output relay contact rating (3.8.11.4)
    - Current: 2 amps continuous (resistive load)
    - Voltage: 28 volts dc
- Power control circuit (3.5.4): Magnetic latch relay Potter and Brumfield Type (SL 11 or HL 11) or as specified in the ordering data

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FSC 1430

Coil rating:
Voltage: 24 vdc (nominal)
Current: 50 ma dc (nominal)
Contact rating:
Voltage: 28 vdc
Current: 2.0 amp (resistive)
Connection: Figure 2
Prime power (3.8.11.1):
Nominal voltage: 28 vdc
Voltage range: 24 to 32 vdc
Input current (3.8.11.2): 130 ma maximum @ 28 vdc (with no channels activated)
Channel activation current: 40 ma maximum per channel at 28 vdc

Performance characteristics:
RF signal level range (3.8.5): +13 dbm to < -110 dbm
Threshold sensitivity (3.8.2): <-110 dbm with a permissible drop out rate of one per channel per five seconds
Response time (3.8.10): 25 milliseconds or less at RF input level ≥ -95 dbm
Multiple command channels (3.8.9): 4

Electromagnetic interference:
Spurious response rejection (antenna) (3.10.1)

	CS04	CS08
Reference level:	-110 dbm	-110 dbm
Limit A:	- 50 dbm	- 50 dbm
Limit B:	- 40 dbm	- 30 dbm (for all frequencies)
except following ranges:		
71 mHz thru 73 mHz	- 40 dbm	0 dbm
225 mHz thru 260 mHz	- 40 dbm	0 dbm
1435 mHz thru 1535 mHz	- 40 dbm	0 dbm
2200 mHz thru 2300 mHz	- 40 dbm	0 dbm
5400 mHz thru 5900 mHz	- 40 dbm	0 dbm

Spurious response (power lead) (3.10.2):
Transient levels: ≤40 volts
Duration: ≤ 2 seconds

Environmental (3.9):
Acceleration level (3.9.1): 100 g
Altitude (3.9.2):

	Condition I	Condition II
Temperature:	-40°C	70°C
Altitude:	≥ .1 in. Hg	≥ .1 in. Hg
Operating time:	30 minutes	30 minutes

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Shock (3.9.3):

Level (g)	Duration (millisecond)
150	1.0
100	10.0

Vibration (3.9.4):

Qualification level	Quality conformance level
Sine curve: S	Sine curve: R
Random curve: AK	Exposure time: 2 min. in each of any 2 axis
Exposure time: 5 min. each axis	

Physical characteristics:

Case (3.11.1): In accordance with figure 1 receiver outline.

Connectors (3.7.2):

J-1: Type "TNC" (Female) RF

J-2: Type MS3112E-12-10SW or PT02E-12-10SW (Bendix Pigmy)

J-3: Type MS3112E-14-19S or PT02E-14-19S (Bendix Pigmy)

J-4: Type MS3112E-12-10S (Bendix Pigmy)

Wiring: In accordance with figure 2 output wiring

Total weight (3.11.1): ≤ 3.5 pounds

TABLE 1. Function table.

Command channel inputs channel designation				Relay operation relay designation			
1st	2nd	3rd	4th	1	2	3	4
0 1/	0	0	0	0	0	0	0
0	0	0	1 2/	0	0	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1
0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	1
1	1	0	0	1	1	0	0
1	1	0	1	1	1	0	1
1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	1

1/ 0 = Off or not activated

2/ 1 = On or activated

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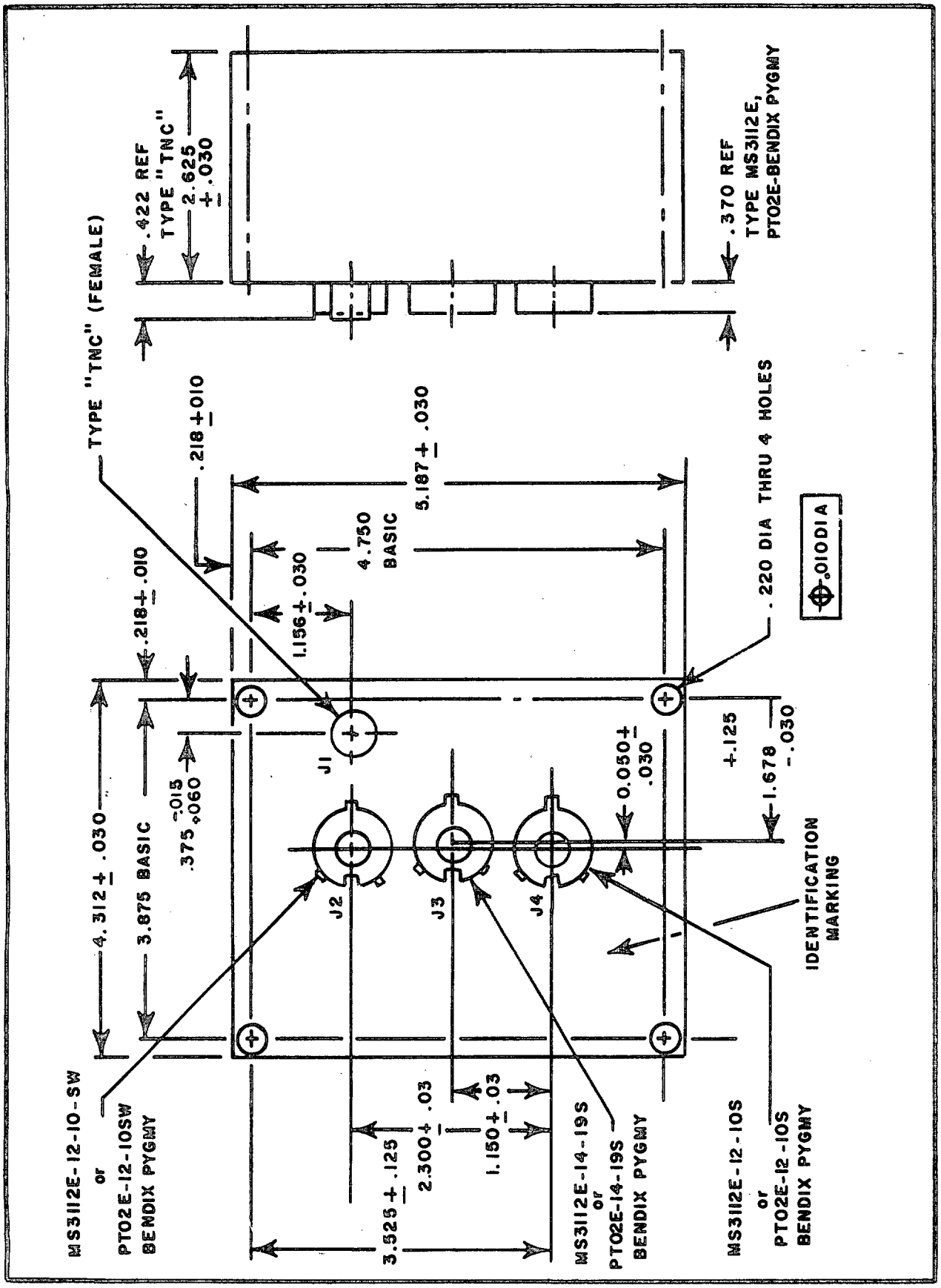


FIGURE 1. RECEIVER OUTLINE

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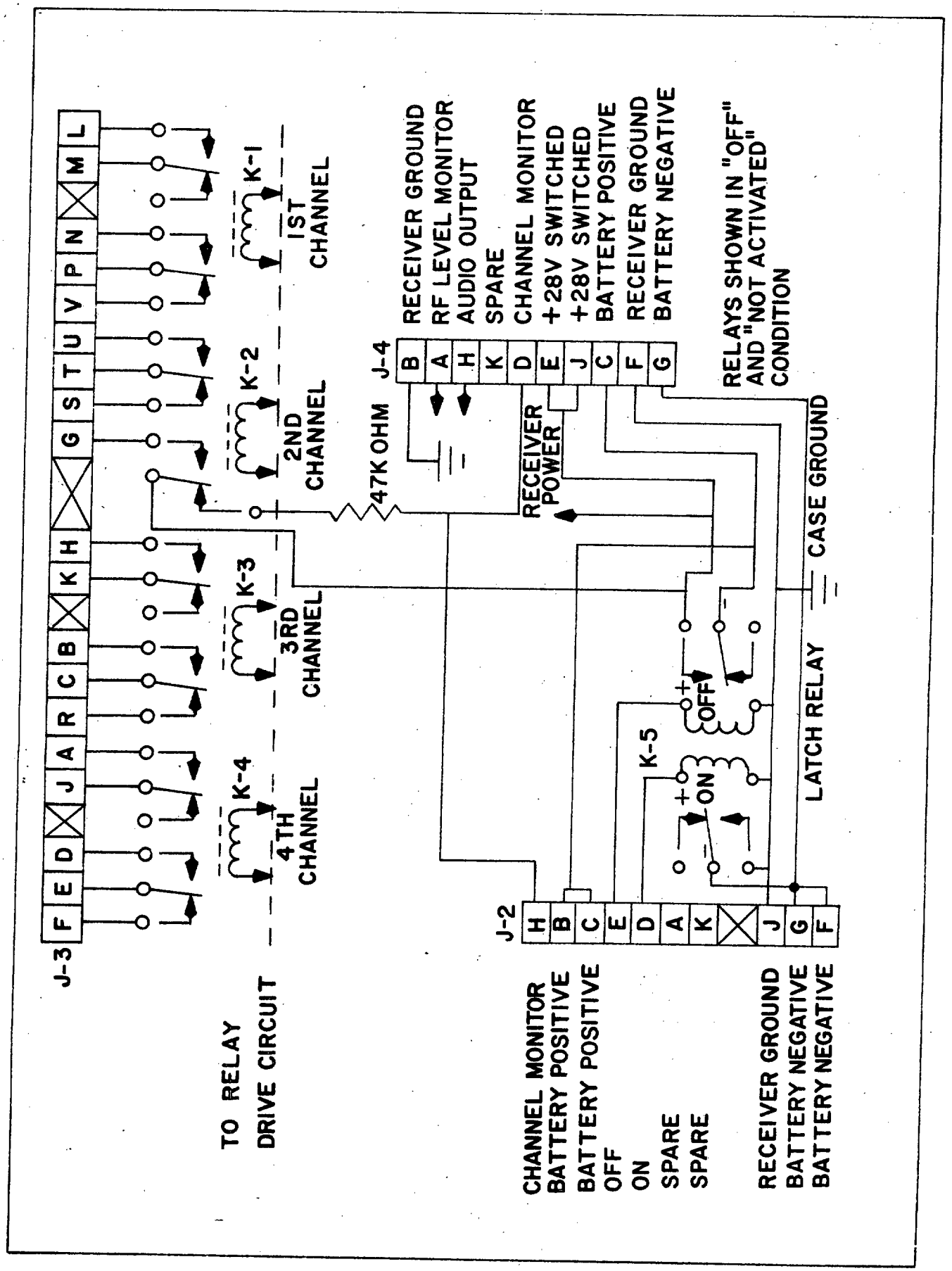


FIGURE 2. CONNECTOR WIRING

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